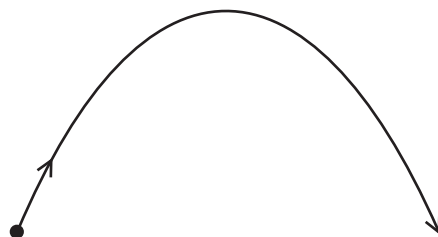


- 8 A stone is thrown upwards and follows a curved path.



Air resistance is negligible.

Why does the path have this shape?

- A The stone has a constant horizontal acceleration and constant vertical velocity.
 - B The stone has a constant horizontal velocity and constant vertical acceleration.
 - C The stone has a constant upward acceleration followed by a constant downward acceleration.
 - D The stone has a constant upward velocity followed by a constant downward velocity.
- 9 A rocket engine ejects 90 kg of exhaust gas per second at a velocity of 190 m s^{-1} relative to the rocket.
- What is the force acting on the rocket due to the ejected gas?
- A 2.1 kN B 17 kN C 18 kN D 162 kN
- 10 Which statement does **not** describe an elastic collision between two objects?
- A The relative speed of approach of the two objects equals the relative speed of separation.
 - B The total kinetic energy of the objects is conserved.
 - C The total kinetic energy of the objects is reduced.
 - D The total linear momentum of the objects is conserved.
- 11 A cyclist is riding at a constant speed on a level road.

According to Newton's third law of motion, what is equal and opposite to the backward push of the back wheel on the road?

- A the force exerted by the cyclist on the pedals
- B the forward push of the road on the back wheel
- C the tension in the cycle chain
- D the total air resistance and friction force